

BUI THI QUYNH NHU

Data Engineering · Business Analytics · AI Engineering

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[Github](#)

[Portfolio](#)

SUMMARY OF QUALIFICATIONS

Computer Science undergraduate at Ton Duc Thang University with 8 end-to-end projects spanning data engineering, cloud infrastructure, SQL-driven analytics, and full-stack systems. Demonstrated ability to build data pipelines and validation workflows in Python and SQL, design and query relational database systems, and architect containerised cloud infrastructure for reliability and performance. Seeking a Data Engineering Internship to contribute to scalable data solutions, ETL workflows, data quality initiatives, and cloud-based analytics platforms.

EDUCATION

Ton Duc Thang University — Ho Chi Minh City

Sep 2023 – Present

Bachelor of Science in Computer Science · GPA: **3.2 / 4**

Relevant coursework: Machine Learning, Digital Image Processing, Database Systems, Data Structures & Algorithms, Cloud Computing, Operating Systems

TECHNICAL SKILLS

Languages & Query: Python (PyTorch · TensorFlow · Pandas · NumPy · OpenCV), SQL, JavaScript, PHP

Data Engineering & Analytics: BigQuery · Data Processing · Data Cleaning · Data Validation · Data Transformation · Data Quality · Data Modeling · ETL Concepts · Statistical Analysis · Feature Engineering

Databases: SQL Databases · Query Optimization · Index Design · Database Design · ERD Design

Cloud & Infrastructure: Docker · AWS · Nginx · Linux · REST APIs · DevOps · Cloud Security · Postman

Web & Tools: Node.js · HTML/CSS · Git · Real-time Monitoring · System Administration

Soft Skills: Analytical Thinking · Technical Communication · Project Management · Cross-functional Teamwork

Business Analysis: Requirement Gathering · Stakeholder Analysis · Use Case Modeling · UML (Sequence, Class) · ERD Design · Business Process Modeling · Agile Scrum · Test Case Design · System Analysis · Technical Documentation

ML / AI: CNN · LSTM · YOLO · Transfer Learning · Feature Engineering · Model Evaluation · Data Augmentation

RESEARCH & PROJECT EXPERIENCE

Sign Language Recognition System

May 2026 – Jun 2026

Computer Vision / Real-Time AI · Sequence Modeling · Role: Data & AI Engineer

- Built an end-to-end data pipeline using MediaPipe hand-landmark extraction to convert raw images and video into **63,915 model-ready sequences** across **38 classes** (26-letter ASL alphabet plus auxiliary and word-level signs), with each sample structured as a 30-frame (**30, 126**) feature tensor.
- Implemented data validation rules to skip frames with no detected hand landmarks and flag classes with insufficient samples, then split data into train/validation/test (**70/15/15 — 44,740 / 9,587 / 9,588 samples**).
- Built and trained a **2-layer Bidirectional GRU sequence model** (~250K parameters) and a real-time inference pipeline recognizing both static and motion-based hand gestures from webcam streams with low latency.
- Identified severe class imbalance between alphabet classes (1,000+ samples each) and word-level signs (under 35 samples each), informing data collection priorities and masking strategy for variable-length sequences.
- Evaluated on held-out test data, achieving **98.5% overall accuracy** and **0.91 macro F1-score**; per-class accuracy reporting and confusion matrix analysis pinpointed which low-sample classes needed further data collection.

Employee Attrition Analytics & Prediction

Mar 2026

Business Analytics Project | Role: Data Analyst

- Performed end-to-end workforce analytics on **14,700 employee records** with **31 predictive features**, including data cleaning, exploratory analysis, feature engineering, and reporting.
- Identified key attrition drivers through statistical analysis, revealing that employees working overtime had a **30.53% attrition rate versus 10.44%** for non-overtime employees, while overall attrition reached 16.12%.
- Built and evaluated **Logistic Regression and Random Forest models** for employee turnover prediction, achieving **ROC-AUC scores of 0.81 and 0.79**, respectively.
- Generated actionable retention insights showing that lower income, shorter tenure (<2 years), and overtime workload were the strongest indicators of employee departure risk.

Minicloud — Cloud Infrastructure Simulator

Oct 2025 – Nov 2025

Systems Engineering · Role: Full-Stack & Cloud Architect

- Architected and deployed a **9-service containerised cloud system** (Docker) replicating AWS compute, storage, and identity management.
- Engineered Nginx intranet routing layer sustaining **zero packet-loss across 100+ simulated concurrent connections** during stress tests.
- Implemented RBAC identity controls and real-time health dashboards, reducing mean incident response time to **under 2 minutes**.

Computer Shop E- Commerce Platform

Aug 2025 – Sep 2025

Full-Stack Web Development · Role: Full-Stack Engineer

- Developed e-commerce platform supporting **500+ SKUs** with dynamic filtering, multi-category management, and end-to-end order processing.
- Optimised SQL query execution by **35%** through index design and query restructuring on high-frequency product search endpoints.
- Implemented paginated REST APIs + caching layer, cutting average page load from **1.8s → 0.7s**.

Real-Time Note-Taking Web Application

Apr 2025 – May 2025

Full-Stack / Cloud Deployment · Role: Full-Stack Engineer

- Deployed cloud-hosted application achieving **99.5% uptime** over a 3-week live testing period with concurrent read/write operations.
- Designed RESTful API backend with optimistic locking, eliminating data conflicts under concurrent access.

Visual Question Answering System

Feb 2026 – Mar 2026

Deep Learning Research Project · Role: Algorithm Designer

- Built a multi-modal data processing pipeline integrating image and text inputs through a 3-component architecture: **ResNet50** (visual feature extractor) + **BiLSTM** (bidirectional question encoder) + **LSTM decoder** (answer generation), achieving multi-class prediction across 3 answer categories.
- Leveraged pretrained ResNet50 ImageNet weights to reduce training time by **~40%** vs. training from scratch, with full benchmark comparison documented.
- Conducted **5-configuration ablation study** comparing BiLSTM vs. LSTM encoders, attention fusion strategies, and dropout rates — identifying optimal architecture.

Traffic Sign Recognition System

Oct 2025 – Nov 2025

Computer Vision / Real-Time AI · Role: Image Processing Lead

- Built a real-time video data processing pipeline (OpenCV + YOLO) achieving **91%+ classification accuracy** across 4 sign categories at 24 fps live stream.
- Extracted **6 colour and 4 geometric features per frame**; integrated object tracking to maintain sign traces across frames.
- Reduced false detections by **25%** via per-class confidence threshold calibration using precision-recall curve analysis.

Fruit Quality & Ripeness Recognition

Nov 2024 – Feb 2025

Undergraduate AI Research · Role: Algorithm Designer & Researcher

- Built a data processing and classification pipeline detecting ripeness across **4 quality grades**, advancing to university competition finals — placing in the **top 15% of submissions**.
- Designed custom augmentation pipeline (rotation, colour jitter, crop) that improved out-of-sample generalisation by **~20%** on held-out test data.
- Reduced misclassification false-positive rate by **18%** through class-weighted loss functions and iterative hyperparameter tuning.

TechSync Store - BA Case Study

Aug 2025 – Sep 2025

Business Analyst

- Conducted user survey and stakeholder analysis to define system requirements
- Designed UML diagrams (Use Case, Sequence, Class) and ERD
- Defined functional & non-functional requirements
- Built test plans and system-level test cases
- Mapped user flow: Browse → Cart → Checkout → Order Management
- Agile Scrum with sprint planning & requirement traceability matrix

PROFESSIONAL EXPERIENCE

Mathematics, Science & English Tutor

Nov 2024 – Jun 2025

eTeacher Company · Ho Chi Minh City

- Delivered **in-home Mathematics and Science tutoring entirely in English**, strengthening both subject fluency and English communication skills for high school students.
- Conducted **online English language classes** via digital platforms, adapting lesson delivery for remote learners across varying proficiency levels.
- Designed structured, step-by-step problem-solving frameworks — analytical skills directly applicable to data pipeline debugging and root-cause analysis.

Programming Instructor

May 2026 – Present

MindX Technology School · Ho Chi Minh City

- Teach Scratch programming and game development fundamentals through interactive, project-based lessons.
- Guide students in logical thinking, debugging, sequencing, conditional logic, loops, and event-driven programming.
- Support students in building simple games while developing problem-solving skills, creativity, and confidence with programming concepts.

CERTIFICATIONS & ACTIVITIES

IELTS 6.5 · Aptis 163 (Advanced English)

2024 – 2025

Member - Technology Club (Programming · AI · Robotics)

2023 – 2025